HHCKLA Buddhist Leung Chik Wai College

6A error analysis (2014)

Student No.: 13 Name: _____ Marks: / 42

Class: 6A Date: _____ Marked date: _____

What have you learned in this chapter? _____

What can you do to improve your work next time? _____

1. If $a > b$ and $k < 0$, which of the following must be true?

   I. $a^2 > b^2$

   II. $a + k > b + k$

   III. $\frac{a}{k^2} > \frac{b}{k^2}$

   A. I only

   B. II only

   C. I and III only

   D. II and III only

(DSE-2014 PII Q06)

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2. There are 792 workers in a factory. If the number of male workers is 20% less than that of female workers, then the number of male workers is

A. 352.

B. 360.

C. 432.

D. 440.

(DSE-2014 PII Q09)

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3. The width and the length of a thin rectangular metal sheet are measured as 8 cm and 10 cm correct to the nearest cm respectively. Let x cm^2 be the actual area of the metal sheet. Find the range of values of x.

A. $71.25 \leq x < 89.25$

B. $71.25 < x \leq 89.25$

C. $79.5 \leq x < 80.5$

D. $79.5 < x \leq 80.5$

(DSE-2014 PII Q11)

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4. It is given that $\frac{4}{5a} = \frac{5}{7b} = \frac{7}{9c}$, where a, b and c are positive numbers. Which of the following is true?

A. $a < b < c$

B. $a < c < b$

C. $b < a < c$

D. $b < c < a$

(DSE-2014 PII Q12)

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5. If z varies inversely as x and directly as the cube of y, which of the following must be constant?

A. xy^3z

B. x^3yz^3

C. $\frac{y^3}{xz}$

D. $\frac{y}{x^3z^3}$

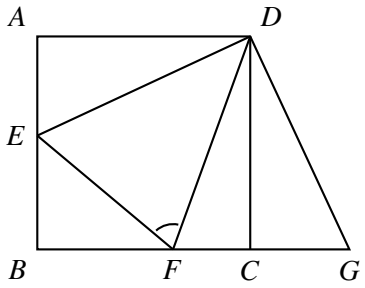
(DSE-2014 PII Q13)

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6. In the figure, $ABCD$ is a square. BC is produced to G such that $\angle CDG = 25°$. E is a point lying on AB such that $AE = CG$. If F is a point lying on BC such that $\angle CDF = 20°$, then $\angle DFE =$



A. 60°.

B. 65°.

C. 70°.

D. 73°.

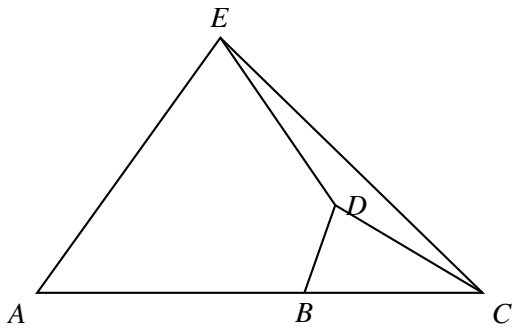
(DSE-2014 PII Q16)

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7. In the figure, B is a point lying on AC such that $AB : BC = 3 : 2$. It is given that $AE \parallel BD$. If the area of $\triangle BCD$ and the area of $\triangle CDE$ are 4 cm^2 and 8 cm^2 respectively, then the area of the trapezium $ABDE$ is



A. 18 cm^2.

B. 21 cm^2.

C. 27 cm^2.

D. 33 cm^2.

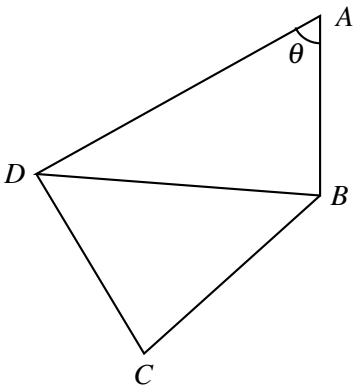
(DSE-2014 PII Q17)

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8. In the figure, $\angle ABD = \angle ADC = \angle BCD = 90°$. If $AB = \ell$, then $CD =$



A. $\ell \sin\theta$.

B. $\ell \cos\theta$.

C. $\ell \sin\theta \tan\theta$.

D. $\dfrac{\ell \tan\theta}{\cos\theta}$.

(DSE-2014 PII Q18)

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9. $\big(\cos(90^\circ+\theta)+1\big)\big(\sin(360^\circ-\theta)-1\big) =$

A. $-\cos^2\theta$.

B. $-\sin^2\theta$.

C. $\cos^2\theta$.

D. $\sin^2\theta$.

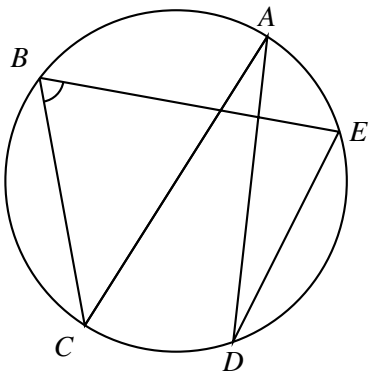
(DSE-2014 PII Q19)

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10. In the figure, AC is a diameter of the circle $ABCDE$. If $\angle ADE = 28°$, then $\angle CBE =$



A. 56°.

B. 62°.

C. 72°.

D. 76°.

(DSE-2014 PII Q20)

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11. The rectangular coordinates of the point P are $(-1, \sqrt{3})$. If P is reflected with respect to the x-axis, then the polar coordinates of its image are

A. $(2, 210°)$.

B. $(2, 240°)$.

C. $(4, 210°)$.

D. $(4, 240°)$.

(DSE-2014 PII Q23)

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12. The stem-and-leaf diagram below shows the distribution of the ages of the passengers in a bus.

Stem (tens)	Leaf (units)
1	h 4 6
2	3 3 3 4 6 7 7
3	1 2 2 2 6 8
4	0 k

If the range of the above distribution is at least 33, which of the following must be true?

I. $0 \leq h \leq 3$

II. $3 \leq k \leq 9$

III. $3 \leq k - h \leq 5$

A. I only

B. II only

C. I and III only

D. II and III only

(DSE-2014 PII Q30)

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13. $7 \times 2^{10} + 2^8 + 5 \times 2^3 - 2^3 =$

A. 111010100000_2.

B. 111100010000_2.

C. 1110100100000_2.

D. 1111000010000_2.

(DSE-2014 PII Q34)

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14. If β is a real number, then $\dfrac{\beta^2+4}{\beta+2i} =$

A. $\beta-2i$.

B. $\beta+2i$.

C. $2-\beta i$.

D. $2+\beta i$.

(DSE-2014 PII Q36)

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15. If $m > 1$, which of the following are geometric sequences?

I. $2^m, 2^{2m}, 2^{3m}, 2^{4m}$

II. $m, 2m^2, 3m^4, 4m^8$

III. $\log m, \log m^2, \log m^4, \log m^8$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

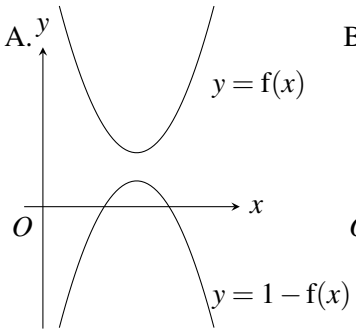
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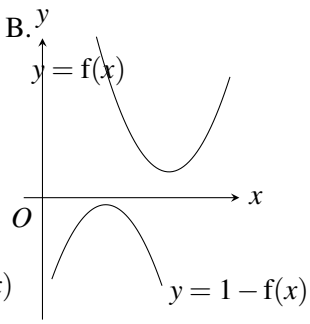
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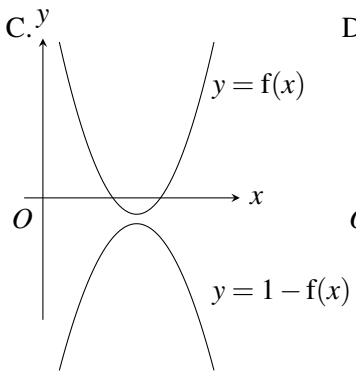
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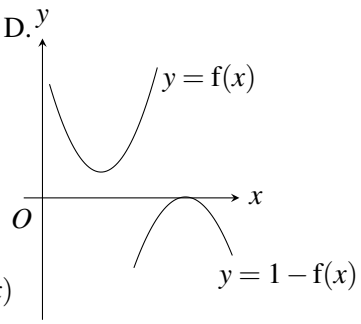
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16. Which of the following may represent the graph of $y = f(x)$ and the graph of $y = 1 - f(x)$ on the same rectangular coordinate system?

A.


B.


C.


D.


(DSE-2014 PII Q38)

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17. For $0° \leq x \leq 360°$, how many roots does the equation $7\sin^2 x = \sin x$ have?

A. 2

B. 3

C. 4

D. 5

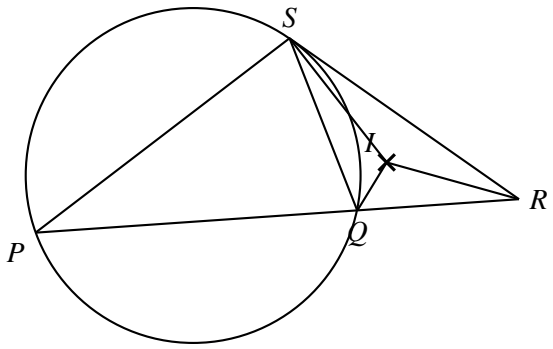
(DSE-2014 PII Q39)

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18. In the figure, PQS is a circle. PQ is produced to R such that RS is the tangent to the circle at S. I is the in-centre of $\triangle QRS$. If $\angle IRQ = 12^\circ$ and $\angle PSQ = 70^\circ$, then $\angle QPS =$



A. 24°.

B. 37°.

C. 43°.

D. 62°.

(DSE-2014 PII Q41)

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19. If the straight line $x - y = k$ and the circle $x^2 + y^2 + 2x - 4y - 1 = 0$ intersect at A and B, then the x-coordinate of the mid-point of AB is

A. $1 + k$.

B. $1 - k$.

C. $\dfrac{1+k}{2}$.

D. $\dfrac{1-k}{2}$.

(DSE-2014 PII Q42)

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20. In an examination, Peter gets 55 marks and his standard score is -3 while Mary gets 95 marks and her standard score is 2. Find the mean of the examination scores.

A. 8 marks

B. 64 marks

C. 75 marks

D. 79 marks

(DSE-2014 PII Q44)

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21. If the variance of the four numbers a, b, c and d is 9, then the variance of the four numbers $14-a$, $14-b$, $14-c$ and $14-d$ is

A. 5.

B. 9.

C. 23.

D. 121.

(DSE-2014 PII Q45)

I can finish _____ questions

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